

Radwa Khattab

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Technical Experience

- **Senior AI Engineer | Proxify – Sweden** **May '26 — now**
 - Led the architecture and development of an AI-powered performance intelligence platform that analyzes the drivers of human and organizational performance.
 - Modernized the platform's backend, infrastructure, and analytics stack, transforming a legacy uncomplete MVP into a scalable cloud-native architecture.
 - Redesigned the database from heavily nested JSONB structures into a normalized PostgreSQL schema (25+ relational tables), significantly improving maintainability, query performance, and scalability.
 - Rebuilt the legacy Ruby analytics engine as a Python-based calculation engine, exposing FastAPI endpoints and deploying it as a Dockerized Azure microservice.
 - Designed and implemented an AI coaching microservice capable of intent detection, evidence retrieval, structured database querying, and personalized analytics generation.
 - Migrated source control from GitHub to Azure DevOps and built CI/CD pipelines for automated validation and deployments across staging and production environments.
 - Designed Azure cloud infrastructure, including containerized services, private networking, private DNS, and secure communication between web applications and backend microservices.
 - Reduced cloud infrastructure costs by approximately **99%** by replacing an over-engineered Databricks-based architecture with a streamlined PostgreSQL solution better aligned with product requirements.
- **Applied & Data Scientist | Microsoft – Egypt** **May '24 — May'26**
 - Contributed to the onboarding of **10M+ product variants** into an Azure ML product-page extraction pipeline to support a hierarchical product model (Product → Variant → Offer), expanding catalog coverage and improving product discovery across Shopping and Copilot experiences.
 - Designed and implemented a fine-tuned Small Language Model (SLM) pipeline on Azure ML, demonstrating the potential to significantly reduce GPT usage and cloud inference costs at scale.
 - Built a cross-team analytics platform on PowerBI using Azure Data Factory pipelines, providing automated insights and debugging capabilities for product managers and engineering teams.
 - Contributed to the Agentic Shopping Platform by integrating multi-agent workflows using A2A communication and developing Semantic Kernel plugins like Other Buying Options support.
 - Developed an automated anomaly detection pipeline for product price updates, improving data quality and increasing the reliability of shopping experiences.
- **AI Engineer | Ejada – Egypt** **Jan. '23 — Jan.'24**
 - Developed AI models for stock market prediction in Saudi Arabia and built most of the interface with NodeJS with the help from a Front-end engineer
 - Worked with the PM and Manager to present the project at a major conference in Saudi Arabia for potential buyers
- **Machine Learning Engineer | Silverkey Technologies – Egypt** **Jan.'20 — Dec.'22**
 - Collaborated with Zewail City and ITIDA on vision transformer research and published a [paper](#) on it.

- Worked on computer vision projects, starting with videos to analytical results obtained from images and flow of motion using OpenCV and DL techniques
- Worked on 3D reconstruction of series of images to build a 3D model on Blender using mathematical approaches

Teaching Experience (Part-time)

- Session Lead | **Udacity** Oct.'22 — Sept.'23
- Teaching Assistant | **Cairo University** Feb.'23 — May'23
- Teaching Assistant | **Nile University** Feb.'22 — Aug.'22
- Machine Learning Instructor | **AMIT Learning** Feb.'22 — Mar.'22

Education

- Master of Science in Informatics, Comp. Engineering, 2022—2026 (Cairo University)
 - **Related Courses:** Advanced NLP, Reinforcement Learning, Data Analytics, Machine Learning, Computer Vision, Advanced Image Processing, Neural Networks, Machine Intelligence 2
 - **GPA:** 3.1
- Bachelor of Science in Computer Engineering, class of 2021 (Cairo University)
 - **Grade:** Excellent
 - **Graduation Project Grade:** Excellent

Published Papers

- [ViTCN: Hybrid Vision Transformer with Temporal Convolution for Multi-Emotion Recognition](#)

Technical Skills

Languages: Python, R, C++, SQL, C#, JavaScript, TypeScript

AI/ML: LLMs & Agents, prompt-engineering, STT evaluation, text-to-structure pipelines; PyTorch, TensorFlow, Keras, OpenCV, Pandas, NumPy, SciPy, PySpark, MLFlow, DVC, CUDA

Backend: FastAPI, Flask, .NET, REST APIs, Pydantic, PostgreSQL; secure API workflows, validation, error handling, rate limiting, E2FA, brute-force protection, SSL-required DB access, Azure Key Vault

DevOps & Infra: Azure (DataFactory, AutoML, Databricks, Foundry, DevOps), Docker, GitHub, GitLab, Azure AML Pipelines

Tools: Linux, PowerBI, Postman, Bruno, Slack, GPT tools; LaTeX, Excel

Extra Curricula Activities

- ICPC 3 years participation (2018—2020)
- WeDo 2.0 (LEGO®) for Juniors (7—12 years kids) Tutor

Language

- **English:** Advanced (TOEFL C1)
- **Arabic:** Native